INCH-POUND

MIL-DTL-85633C
15 January 2005
SUPERSEDING
MIL-DTL-85633B
26 February 2001

DETAIL SPECIFICATION

# COVERALLS, FLYERS', ANTI-EXPOSURE, CWU-62B/P, CWU-74/P, CWU-62C/P, CWU-86/P, AND CWU-87/P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers five types of continuous wear anti-exposure flyers' coveralls designated as CWU-62B/P, CWU-74/P, CWU-62C/P, CWU-86/P and CWU-87/P.

1.2 Classification. The coveralls will be one of the following types and sizes as specified (see 6.2).

1.2.1 Types. The types of coveralls are as follows:

Type I - CWU-62B/P, without pockets, wrist seals, and socks, Naval Air Systems Command application, men's sizing.

Type II - CWU-74/P, with pockets, wrist seals, and socks, US Air Force application, men's sizing.

Type III - CWU-62C/P, without pockets, wrist seals, and socks, Naval Air Systems Command application, women's sizing.

Type IV - CWU-86/P, ruggedized with overvest, pockets, neck seals, and wrist seals, without socks, Naval Air Systems Command application, men's sizing.

Type V - CWU-87/P, ruggedized with overvest, pockets, neck seals, and wrist seals, without socks, Naval Air Systems Command application, women's sizing.

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division (Code 491000B120-3), Highway 547, Lakehurst, NJ 08733-5100 or emailed to Thomas.Omara@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 8475

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1.2.2 Sizes. The anti-exposure coverall sizes are as follows:

Types I, II, and IV (Men's Sizing):

| | | |
|---|---|---|
| Size 1 (32-36) | - | Small-Short |
| Size 2 (32-36) | - | Small-Regular |
| Size 3 (32-36) | - | Small-Long |
| Size 4 (37-40) | - | Medium-Short |
| Size 5 (37-40) | - | Medium-Regular |
| Size 6 (37-40) | - | Medium-Long |
| Size 7 (41-44) | - | Large-Short |
| Size 8 (41-44) | - | Large-Regular |
| Size 9 (41-44) | - | Large-Long |
| Size 10 (45-47) | - | Extra Large-Short |
| Size 11 (45-47) | - | Extra Large-Regular |
| Size 12 (45-47) | - | Extra Large-Long |

Types III and V (Women's Sizing):

| | | |
|---|---|---|
| XSmall | - | Short |
| XSmall | - | Regular |
| XSmall | - | Long |
| XSmall | - | XLong |
| Small | - | Short |
| Small | - | Regular |
| Small | - | Long |
| Small | - | XLong |
| Medium | - | Short |
| Medium | - | Regular |
| Medium | - | Long |
| Medium | - | XLong |
| Large | - | Short |
| Large | - | Regular |
| Large | - | Long |
| Large | - | XLong |
| Xlarge | - | Short |
| Xlarge | - | Regular |
| XLarge | - | Long |
| XLarge | - | XLong |
| XXLarge | - | Short |
| XXLarge | - | Regular |
| XXLarge | - | Long |
| XXLarge | - | XLong |



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## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

COMMERCIAL ITEM DESCRIPTIONS

| | | |
|---|---|---|
| A-A-50195 | - | Thread, Aramid |
| A-A-55126 | - | Fastener Tapes, Hook and Loop, Synthetic |
| A-A-55634 | - | Zippers (Fasteners, Slide Interlocking) |

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | | |
|---|---|---|
| MIL-DTL-32075 | - | Label: For Clothing, Equipage, and Tentage, (General Use) |
| MIL-DTL-32149 | - | Cloth, Waterproof and Moisture Vapor Permeable, Flame Resistant |
| MIL-C-83429 | - | Cloth, Plain and Basket Weave, Aramid |
| MIL-C-85101 | - | Cloth, Warp Knit, Aramid, High Temperature Resistant |
| MIL-W-85635 | - | Water Test Device, Anti-Exposure Coverall |
| MIL-C-85637 | - | Cloth, Laminated, Fire Resistant, Waterproof and Moisture Vapor Permeable |

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.



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Type I (CWU-62B/P COVERALLS)

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| 1370AS502 | - | Seam Tape and Patches |
| 1725AS102 | - | Neck Seal, Anti-Exposure Coverall, CWU-62B/P |
| 3324AS103 | - | Coveralls Subassembly, Anti-Exposure, Flyers', Basic, PTFE, Constant Wear |
| 3769AS103 | - | Zipper, Omni-Environment Barrier |

Type II (CWU-74/P COVERALLS)

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| 1370AS403 | - | Coveralls, Flyers', Anti-Exposure, CWU-74/P |
| 1370AS404 | - | Coveralls, Subassembly, Anti-Exposure, Flyers' PTFE, Constant Wear |
| 1370AS502 | - | Seam Tape and Patches, Heat Sealable Anti-Exposure Coverall, CWU-74/P |
| 1370AS706 | - | Wrist and Neck Seal Pattern, Sizes 1 thru 12, CWU-74/P Anti-Exposure Coverall |
| 3769AS103 | - | Zipper, Omni-Environment Barrier |

Type III (CWU-62C/P COVERALLS)

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| 1370AS502 | - | Seam Tape and Patches |
| 1725AS102 | - | Neck Seal, Anti-Exposure Coverall, CWU-62B/P |
| 3653AS100 | - | Coveralls, Anti-Exposure, CWU-62C/P |
| 3769AS103 | - | Zipper, Omni-Environment Barrier |

Type IV (CWU-86/P COVERALLS)

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| 1370AS502 | - | Seam Tape and Patches |
| 3769AS100 | - | Coveralls Assembly, Flyers' Anti-Exposure, Ruggedized, CWU-86/P |
| 3769AS101 | - | Coveralls, Flyers' Anti-Exposure, Ruggedized, CWU-86/P |
| 3769AS102 | - | Overvest Subassembly, Flyers' Anti-Exposure, Ruggedized, CWU-86/P |



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| | | |
|---|---|---|
| 3769AS103 | - | Zipper, Omni-Environment Barrier, Anti-Exposure |
| 3769AS105 | - | Pockets, Subassembly, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |
| 3769AS110 | - | Seal, Neck, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |
| 3769AS111 | - | Seal, Neck, Gasket Anti-Exposure, CWU-86/P & CWU-87/P |
| 3769AS120 | - | Seal, Wrist, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |

Type V (CWU-87/P COVERALLS)

NAVAL AIR SYSTEMS COMMAND

| | | |
|---|---|---|
| 1370AS502 | - | Seam Tape and Patches |
| 3769AS103 | - | Zipper, Omni-Environment Barrier, Anti-Exposure |
| 3769AS105 | - | Pockets, Subassembly, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |
| 3769AS110 | - | Seal, Neck, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |
| 3769AS111 | - | Seal, Neck, Gasket Anti-Exposure, CWU-86/P & CWU-87/P |
| 3769AS120 | - | Seal, Wrist, Flyers' Anti-Exposure, Ruggedized, CWU-86/P & CWU-87/P |
| 3769AS200 | - | Coveralls Assembly, Flyers' Anti-Exposure, Ruggedized, CWU-87P |
| 3769AS201 | - | Coveralls, Flyers' Anti-Exposure, Ruggedized, CWU-87P |
| 3769AS202 | - | Overvest Subassembly, Coveralls, Flyers' Anti-Exposure, Ruggedized, CWU-87/P |

(Copies of these drawing are available online at http://ct.dscp.dla.mil/contracting/, or from Defense Supply Center Philadelphia, Code CBTC Bldg. 6, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

| | | |
|---|---|---|
| ASQ-Z1.4 | - | Procedures, Sampling and Tables for Inspection by Attributes (DoD adopted) |

(Copies of this document are available from www.asq.org or American Society for Quality, 600 Plankinton Avenue, Milwaukee, WI 53203.)



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AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM) INTERNATIONAL

ASTM-D6193 - Stitches and Seams, Standard Practice for (DoD adopted)

(Copies of these documents are available from www.atsm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

NATIONAL AEROSPACE STANDARDS COMMITTEE (NASC)

NASM43770-1 - Snap Hooks, Wire Body, Fixed Loop Eye, Flat Spring Closure, with Retainer

(Copies of this document are available from www.aia-aerospace.org or Aerospace Industries Association, 1000 Wilson Blvd, Suite 1700, Arlington, VA 22209-3901.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 Qualification. The anti-exposure coveralls furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3). Qualification of any one type and size of coverall under this specification shall automatically qualify a supplier for all types and all sizes specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Materials and components. Unless otherwise specified, the materials and components shall have been manufactured not greater than eighteen months prior to the date of delivery of the anti-exposure coverall.

3.4.1 Laminated cloth (Types I, II and III). The base cloth for types I, II, and III coveralls shall be a three layer laminated package conforming to MIL-C-85637.



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3.4.2 Laminated cloth (Types IV and V). The base cloth for types IV and V coveralls shall conform to MIL-DTL-32149.

3.4.3 Raschel knit cloth. The knit cloth for the expansion panel shall be a high temperature resistant aramid, two-bar raschel warp knit conforming to MIL-C-85101. The color shall be Sage Green 1565 (see 4.5.1).

3.4.4 Aramid cloth (Types IV and V). The aramid cloth for the overvest, pockets, and pocket tabs shall be a flame resistant, high strength aramid fiber blend conforming to MIL-C-83429, type II, class 6. The color shall be Sage Green 1590.

3.4.5 Heat seal cloth (Types IV and V). The heat seal cloth for the seat and knee reinforcement shall conform to Drawing 1370AS502, part number 1370AS502-5.

3.4.6 Zippers.

3.4.6.1 Entry and portal zippers. The entry zipper and portal zipper, as applicable, shall be an omni-environmental type conforming to Drawing 3769AS103 (see 6.6.1).

3.4.6.2 Zipper thong or tether.

3.4.6.2.1 Zipper thong (Type III). The entry zipper shall have a thong attached. The thong shall be cut from the laminated cloth specified in 3.4.1. The ends shall be finished to prevent raveling or fraying and shall finish $^{3}/_{8}$ to $^{1}/_{2}$ inch wide and 42 inches long ± $^{1}/_{4}$ inch when attached to the zipper pull.

3.4.6.2.2 Zipper thong and tether (Type V).

3.4.6.2.2.1 Zipper thong (Type V). The entry zipper shall have an attached thong conforming to Drawing 3769AS101.

3.4.6.2.2.2 Detachable tether (Type V). The entry zipper shall have a detachable tether. The tether shall be cut from aramid cloth specified in 3.4.4. One end shall be attached to a spring closure snap hook as specified in 3.4.9. The other end of the cloth shall be finished to prevent raveling or fraying. The tether shall finish $^{3}/_{8}$ to $^{1}/_{2}$ inch wide and 42 inches long ± $^{1}/_{4}$ inch when attached to the zipper pull.

3.4.6.3 Pockets and overvest zippers (Types IV and V).



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3.4.6.3.1 Pencil pocket. The zipper for the pencil pocket shall conform to A-A-55634, type I, style 7, 6-1/2 inches long (see 6.6.2).

3.4.6.3.2 Calf pocket. The zipper for the calf pocket shall conform to A-A-55634, type I, style 4, 9 inches long (see 6.6.2).

3.4.6.3.3 Overvest zipper. The zipper for the overvest shall conform to A-A-55634, type I, style 4 (see 6.6.2). The zipper lengths shall be as follows:

| Men's Size | Type IV | Women's Size | Type V |
|---|---|---|---|
| S | 8 inches | XS | 8 inches |
| M | 8 inches | S | 8 inches |
| L | 9 inches | M | 9 inches |
| XL | 9 inches | L | 9 inches |
| | | XL | 10 inches |
| | | XXL | 11 inches |

3.4.7 Fastener tapes (Types III, IV, and V). The hook and loop fastener tape shall be $^{3}/_{4}$ inch wide and 2 inches wide and shall conform to A-A-55126, type II, class 1. The color of the hook and loop tape shall match the laminated cloth.

3.4.8 Thread. The thread for stitching the coverall shall be a 60 tex, aramid thread conforming to A-A-50195. The color shall match the appropriate laminated cloth in 3.4.1 or 3.4.2.

3.4.9 Snap hook. The snap hook for the type V tether shall be in accordance with NASM43770-1.

3.5 Labels.

3.5.1 Barcode label. Each coverall shall have a barcode label conforming to MIL-DTL-32075, type VII, class 17. Barcode labels shall not be attached to the coverall in a manner that puts a hole into the material that compromises the waterproof integrity of the coverall or by using an adhesive that leaves a residue upon removal. The label shall be located so that it is completely visible on the item when it is folded or packaged (see 3.15).

3.5.2 Identification label. The coveralls shall have a combination size, identification, and instruction label conforming to MIL-DTL-32075, type VI, class 14. The label can be in one or two pieces. If two pieces are used the warning and caution statements shall be placed on the second label and the second label shall be located directly below the first label. The label shall also include the following information in 10 point type (see 6.7):



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CWU-XX/P (XX shall either be 62B for Type I or 74 for Type II or 62C for Type III or 86 for Type IV or 87 for Type V)
Anti-Exposure Flyers Coverall (For Types I, II, and III)
Anti-Exposure Flyers Ruggedized Coveralls (For Types IV and V)
Date of manufacture (month/year)
Serial Number (see 3.5.3)

CAUTION. (10 point). Machine wash cold with powdered detergent. Drip or tumble dry low.

DO NOT DRY CLEAN

REMOVE WATCHES AND JEWELRY BEFORE DONNING AND DOFFING. Coverall cloth can be damaged. Sharp instruments can puncture or abrade surfaces. Use care when donning or doffing. Ensure zippers are securely closed after donning.

3.5.3 Serial numbers. The coveralls shall be identified by individual serial numbers which shall be assigned by the manufacturer. Serialization shall be by a block of consecutive numbers to cover the entire acquisition document quantity.

3.6 Design.

3.6.1 Type I (CWU-62B/P). The design of the type I coveralls shall conform to the requirements of applicable drawings (see 2.2.2). The coveralls shall be one piece garment containing a neck seal and an omni-environmental barrier entry and a portal zipper (see figure 1).

3.6.2 Type II (CWU-74/P). The design of the type II coveralls shall conform to the requirements of applicable drawings (see 2.2.2). The coveralls shall be a one piece garment containing a neck seal, wrist seals, pockets, socks, and an omni-environmental barrier entry and a portal zipper (see figure 2).

3.6.3 Type III (CWU-62C/P). The design of the type III coveralls shall conform to the requirements of applicable drawings (see 2.2.2). The coveralls shall be a one piece garment containing a neck seal and omni-environmental barrier entry zipper. The zipper shall have a thong to enable the wearer to pull it open and close up and down the back (see figures 3 and 4).

3.6.4 Type IV (CWU-86/P). The design of the type IV coveralls shall conform to the requirements of the applicable drawings (see 2.2.2). The coverall shall be a one piece garment containing a neck seal, wrist seals, an omni-environmental barrier entry and a portal zipper, overvest, and pockets (see figures 5 and 6).



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3.6.5 Type V (CWU-87/P). The design of the type V coveralls shall conform to the requirements of the applicable drawings (see 2.2.2). The coverall shall be a one piece garment containing a neck seal, wrist seals, omni-environmental barrier entry zipper, overvest, and pockets (see figures 7 and 8).

3.7 Construction. The coveralls shall be constructed in accordance with Drawings 3324AS103 for Type I; 1370AS403 and 1370AS404 for Type II; 3653AS100 for Type III; 3769AS100 for Type IV; 3769AS200 for Type V using sewn seams covered on the inside of the coverall by heat sealed tapes.

3.8 Patterns. The Government furnished patterns for use in cutting working patterns (see 6.5) shall not be altered in any way. The Government furnished patterns shall be used as a guide for making the contractor's working patterns. Minor modifications of the contractor's working patterns are permitted when using automated equipment or to accommodate a manufacturer's process but the alterations shall not affect the serviceability, dimensions, or appearance of the coverall.

3.8.1 List of pattern parts. The pattern parts and cutting quantities for the coverall shall be as specified in table I (see 6.5.1).

TABLE I. List of pattern parts.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| I | Laminated cloth | Front Torso | FRT_TORSO_I | 1 |
| | Laminated cloth | Back Torso | BCK_TORSO_I | 1 |
| | Laminated cloth | Top Sleeve | TOPSLEEVE_I | 2 |
| | Laminated cloth | Under Sleeve | UDRSLEEVE_I | 2 |
| | Laminated cloth | Front Leg | FRONT_LEG_I | 2 |
| | Laminated cloth | Back Leg | BACK_LEG_I | 2 |
| | Raschel | Expansion Panel | EXPNSNPNL_I | 2 |
| II | Laminated cloth | Back Torso | -- | 1 |
| | Laminated cloth | Top Sleeve | -- | 2 |
| | Laminated cloth | Under Sleeve | -- | 2 |



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TABLE I. List of pattern parts - Continued.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| II Continued | Laminated cloth | Back Leg | -- | 2 |
| | Laminated cloth | Front Torso | -- | 2 |
| | Laminated cloth | Pencil Pocket Flap | -- | 1 |
| | Laminated cloth | Breast Pocket | -- | 2 |
| | Laminated cloth | Pencil Pocket | -- | 1 |
| | Laminated cloth | Right Thigh Pocket | -- | 1 |
| | Laminated cloth | Lower Leg Pocket | -- | 2 |
| | Laminated cloth | Breast Pocket Cover | -- | 2 |
| | Laminated cloth | Lower Thigh Pocket Cover | -- | 1 |
| | Laminated cloth | Right Leg Pocket Cover | -- | 2 |
| | Raschel | Expansion Panel | -- | 1 |
| III | Laminated cloth | Front Top | FRT_TOP_III | 1 |
| | Laminated cloth | Back Top | BK_TOP_III | 1 |
| | Laminated cloth | Sleeve | SLEEVE_III | 2 |
| | Laminated cloth | Front Leg | FRT_LEG_III | 2 |
| | Laminated cloth | Back Leg | BK_LEG_III | 2 |
| | Laminated cloth | Gusset Inner | GSSTINR_III | 2 |
| | Laminated cloth | Pocket | POCKET_III | 1 |
| | Raschel | Gusset Outer | GSSTOTR_III | 2 |



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TABLE I. List of pattern parts - Continued.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| IV | Laminated cloth | Front | FRONT_IV | 2 |
| | Laminated cloth | Back Torso | BK_TORSO_IV | 1 |
| | Laminated cloth | Back Leg | BACK_LEG_IV | 2 |
| | Laminated cloth | Sleeve | SLEEVE_IV | 2 |
| | Laminated cloth | Sock Attachment Panel | SOCKAHPL_IV | 2 |
| | Laminated cloth | Knife Pocket Reinforcement | KNIFEPKTRNF | 1 |
| | Aramid | Collar | COLLAR_IV | 1 |
| | Aramid | Collar Interlining | CLLRINT_IV | 2 |
| | Aramid | Knife Pocket Backing | KPKTBKNG_IV | 1 |
| | Aramid | Overvest Back | OV_BACK_IV | 1 |
| | Aramid | Overvest Front Facing | OV_FRTFC_IV | 2 |
| | Aramid | Overvest Flap | OV_FLAP_IV | 1 |
| | Aramid | Overvest Front Left | OVLFRONT_IV | 1 |
| | Aramid | Overvest Front Right | OVRFRONT_IV | 1 |
| | Aramid | Yoke | OV_YOKE_IV | 2 |
| | Aramid | Calf Pocket | CALF_PKT | 2 |
| | Aramid | Calf Pocket Upper | CALF_PKTUPR | 2 |
| | Aramid | Glove Pocket | GLOVE_PKT | 2 |



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TABLE I. List of pattern parts - Continued.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| IV Continued | Aramid | Knife Pocket | KNIFE_PKT | 1 |
| | Aramid | Sleeve Cuff | SLEEVE_CUFF | 2 |
| | Aramid | Pencil Pocket Flap | PEN_PKT_FLP | 1 |
| | Aramid | Left Sleeve Pocket | LSLEEVE_PKT | 1 |
| | Aramid | Pencil Pocket | PENCIL_PKT | 1 |
| | Aramid | Left Sleeve Pocket Side | LSLV_PKTSDE | 1 |
| | Aramid | Cuff Adjustment Tab | CUFF_ADJTAB | 2 |
| | Aramid | Thong | THONG | 1 |
| | Aramid | Pocket tab | PKTTAB | 6 |
| | Raschel | Back Expansion Panel | BKEXPPNL_IV | 1 |
| | Heat Seal cloth | Knee Reinforcement Panel | KNEERFPL_IV | 2 |
| | Heat Seal cloth | Seat Reinforcement Panel | SEATRFPL_IV | 2 |
| | Guide | Undercollar Stitching | G_UCLRST_IV | 0 |
| | Guide | Left Front | G_LFRONT_IV | 0 |
| | Guide | Right Front | G_RFRONT_IV | 0 |
| | Guide | Left Sleeve | G_L_SLV_IV | 0 |
| | Guide | Right Sleeve | G_R_SLV_IV | 0 |
| V | Laminated cloth | Front | FRONT_V | 2 |



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TABLE I. List of pattern parts - Continued.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| V Continued | Laminated cloth | Back Top | BACK_TOP_V | 1 |
| | Laminated cloth | Back Leg | BACK_LEG_V | 2 |
| | Laminated cloth | Sleeve | SLEEVE_V | 2 |
| | Laminated cloth | Sock Attachment Panel | SOCKATHPL_V | 2 |
| | Laminated cloth | Knife Pocket Reinforcement | KNIFEPKTRNF | 1 |
| | Laminated cloth | Gusset Inner | GUSSETINR_V | 2 |
| | Aramid | Collar | COLLAR_V | 1 |
| | Aramid | Collar Interlining | CLLR_INT_V | 2 |
| | Aramid | Knife Pocket Backing | KPKTBKNG_V | 1 |
| | Aramid | Overvest Back Left | OV_BACK_L_V | 1 |
| | Aramid | Overvest Back Right | OV_BACK_R_V | 1 |
| | Aramid | Overvest Facing | OV_FRT_FC_V | 2 |
| | Aramid | Overvest Flap | OV_FLAP_V | 2 |
| | Aramid | Overvest Front Left | OV_FRT_L_V | 1 |
| | Aramid | Overvest Front Right | OV_FRT_R_V | 1 |
| | Aramid | OvervestYoke | OV_YOKE_V | 2 |
| | Aramid | Calf Pocket | CALF_PKT | 2 |
| | Aramid | Calf Pocket Upper | CALF_PKTUPR | 2 |



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TABLE I. List of pattern parts - Continued.

| COVERALL TYPE | MATERIAL | PATTERN NOMENCLATURE | COMPUTER NOMENCLATURE | CUT PARTS |
|---|---|---|---|---|
| V Continued | Aramid | Glove Pocket | GLOVE_PKT | 2 |
| | Aramid | Knife Pocket | KNIFE_PKT | 1 |
| | Aramid | Sleeve Cuff | SLEEVE_CUFF | 2 |
| | Aramid | Pencil Pocket Flap | PEN_PKT_FLP | 1 |
| | Aramid | Left Sleeve Pocket | LSLEEVE_PKT | 1 |
| | Aramid | Pencil Pocket | PENCIL_PKT | 1 |
| | Aramid | Left Sleeve Pocket Side | LSLV_PKTSDE | 1 |
| | Aramid | Cuff Adjustment Tab | CUFF_ADJTAB | 2 |
| | Aramid | Thong | THONG | 1 |
| | Aramid | Pocket tab | PKTTAB | 6 |
| | Raschel | Gusset Outer | GUSSETOTR_V | 2 |
| | Heat Seal cloth | Knee Reinforcement Panel | KNEERFPL_V | 2 |
| | Heat Seal cloth | Seat Reinforcement Panel | SEATRNFPL_V | 2 |
| | Guide | Undercollar Stitching | G_UCLRSTH_V | 0 |
| | Guide | Left Front | G_LFRONT_V | 0 |
| | Guide | Right Front | G_RFRONT_V | 0 |
| | Guide | Left Sleeve | G_L_SLV_V | 0 |
| | Guide | Right Sleeve | G_R_SLV_V | 0 |



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3.9 Cutting. All fabric pieces shall be cut in accordance with the patterns. If no directional line is given on the pattern the long dimension of the pieces shall be cut in the direction of the fabric warp.

3.10 Marking. Cut parts shall be marked, ticketed, or bundled to ensure a uniform size throughout the coverall. Any method of marking shall be used except metal fastening devices, sewn on tickets, or adhesive type tickets that leave traces of adhesive on the material after removal of the ticket. Drill holes may be used to mark the lay only in areas where the drill marks will be buried within a seam and covered by seam tape.

3.11 Stitches, seams, and stitching. ASTM-D6193 stitch types 301 and 401, with 8 to 10 stitches per inch shall be used in the construction of the coveralls. Seams shall be as specified in construction drawings (see 3.7).

3.12 Heat sealing. All sewn seams and stitching shall be covered on the inside of the coverall and sealed with the seam tape. The entire width of the seam tape shall be heat sealed over the seams and stitching. Crossover areas and ends left unsealed after heat sealing shall be taped. To maintain a tape overlap, sealed sewn seams shall not contain more than two layers of seam tape per seam. The tape shall not lift from the seam at any point. The neck and wrist seals attaching seams to the coverall shall be covered on the inside of the coveralls. End joints where the tape passes over taped seams on the coveralls shall be avoided. The overlap of splicings and tape ends shall be not less than ¾ inch.

3.12.1 Patches for mending. The coveralls shall contain no more than three mending patches. Patching of fabrication defects shall be permitted by a 2 inch diameter patch (see Drawing 1370AS502) on the inside of the coveralls and of a color to match the color of the backing cloth. Patches shall be applied at a temperature of 350 °F for 20 seconds at 75 ± 2 ft. lbs. pressure. When patching is used it shall be noted on the inspection form (see figures 9, 10, 11, and 12). The patched area shall not leak (see 3.14).

3.13 Finished garment dimensions. The finished garment dimensions shall be as specified in tables II through VI, as applicable. The garment shall lie flat with the zipper fully closed when measuring. Locations A through N are shown on figures 1 through 8. Do not stretch knit at any time during garment measurement.



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TABLE II. Type I - CWU-62B/P finished garment dimensions (inches).

| SIZE | CHEST WIDTH A | SLEEVE LENGTH B | FRONT TORSO LENGTH C | INSEAM LENGTH D |
|---|---|---|---|---|
| 1 (32-36) | 21 ¼ | 20 | 27 ½ | 27 |
| 2 (32-36) | 21 ¼ | 21 ½ | 29 ½ | 29 ½ |
| 3 (32-36) | 21 ¼ | 22 ½ | 31 | 31 |
| 4 (37-40) | 23 | 20 | 27 ½ | 27 |
| 5 (37-40) | 23 | 21 ½ | 29 ½ | 29 ½ |
| 6 (37-40) | 23 | 22 ½ | 31 | 31 |
| 7 (41-44) | 25 | 20 | 27 ½ | 29 |
| 8 (41-44) | 25 | 21 ½ | 29 ½ | 29 ½ |
| 9 (41-44) | 25 | 22 ½ | 31 | 31 |
| 10 (45-47) | 27 ½ | 20 | 27 ½ | 29 |
| 11 (45-47) | 27 ½ | 21 ½ | 29 ½ | 29 ½ |
| 12 (45-47) | 27 ½ | 22 ½ | 31 | 31 |
| Tolerance | ± 1 | ± 5/8 | ± 1 | ± 1 |

A - The chest width shall be measured above the zipper from edge to edge at the underarm seam.

B - The sleeve length shall be measured along the top seam of the back sleeve from the armscye seam to the edge of the sleeve.

C - The front torso length shall be measured along the center front seam from the edge of the neck seal seam to the edge of the material at the crotch. The measuring tape shall be placed to the right side of the relief portal zipper.

D - The inseam shall be measured along the seam from the edge of the crotch seam to the bottom edge of the leg.

TABLE III. Type II - CWU-74/P finished garment dimensions (inches).

| SIZE | CHEST WIDTH A | SLEEVE LENGTH B | FRONT TORSO LENGTH C | INSEAM LENGTH D |
|---|---|---|---|---|
| 1 (32-36) | 14 7/8 | 19 3/8 | 26 | 27 |
| 2 (32-36) | 15 | 20 3/8 | 26 5/8 | 28 5/8 |
| 3 (32-36) | 15 3/8 | 21 7/8 | 28 ¼ | 31 |
| 4 (37-40) | 17 ¼ | 20 1/8 | 28 7/8 | 27 |
| 5 (37-40) | 17 3/8 | 20 7/8 | 29 ¾ | 28 5/8 |
| 6 (37-40) | 17 ½ | 21 7/8 | 30 | 31 |
| 7 (41-44) | 18 ¾ | 20 3/8 | 28 1/8 | 27 |



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TABLE III. Type II - CWU-74/P finished garment dimensions (inches) - Continued.

| SIZE | CHEST WIDTH A | SLEEVE LENGTH B | FRONT TORSO LENGTH C | INSEAM LENGTH D |
|---|---|---|---|---|
| 8 (41-44) | 19 3/8 | 21 3/8 | 30 ¾ | 28 5/8 |
| 9 (41-44) | 19 5/8 | 22 | 31 | 31 |
| 10 (45-47) | 21 ¼ | 19 ½ | 29 ¾ | 27 |
| 11 (45-47) | 21 3/8 | 21 | 30 1/8 | 28 5/8 |
| 12 (45-47) | 21 5/8 | 22 ¼ | 32 | 31 |
| Tolerance | ± 1 | ± 5/8 | ± 1 | ± 1 |

A - The chest width shall be measured above the zipper from edge to edge at the underarm seam.

B - The sleeve length shall be measured along the top seam of the back sleeve from the armscye seam to the edge of the sleeve.

C - The front torso length shall be measured along the center front seam from the edge of the neck seal seam to the edge of the material at the crotch. The measuring tape shall be placed to the right side of the relief portal zipper.

D - The inseam shall be measured along the seam from the edge of the crotch seam to the bottom edge of the leg.

TABLE IV. Type III - CWU-62C/P finished garment dimensions (inches).

| SIZE | CHEST WIDTH E | HIP F | CENTER BACK LENGTH G | SLEEVE LENGTH H | INSEAM LENGTH J |
|---|---|---|---|---|---|
| X-Small / Short | 13 11/16 | 16 ½ | 16 | 23 ½ | 26 |
| X-Small / Regular | 13 11/16 | 16 ½ | 17 | 25 | 28 |
| X-Small / Long | 13 11/16 | 16 ½ | 18 | 26 ½ | 30 |
| X-Small / X-Long | 13 11/16 | 16 ½ | 19 | 28 | 32 |
| Small / Short | 14 7/16 | 18 ½ | 17 | 24 | 26 |
| Small / Regular | 14 7/16 | 18 ½ | 18 | 25 ½ | 28 |
| Small / Long | 14 7/16 | 18 ½ | 19 | 27 | 30 |
| Small / X-Long | 14 7/16 | 18 ½ | 20 | 28 ½ | 32 |



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TABLE IV. Type III - CWU-62C/P finished garment dimensions (inches) - Continued.

| SIZE | CHEST WIDTH E | HIP F | CENTER BACK LENGTH G | SLEEVE LENGTH H | INSEAM LENGTH J |
|---|---|---|---|---|---|
| Medium / Short | 16 3/16 | 20 ½ | 18 | 24 ½ | 26 |
| Medium / Regular | 16 3/16 | 20 ½ | 19 | 26 | 28 |
| Medium / Long | 16 3/16 | 20 ½ | 20 | 27 ½ | 30 |
| Medium / X-Long | 16 3/16 | 20 ½ | 21 | 29 | 32 |
| Large / Short | 17 15/16 | 22 ½ | 19 | 25 | 26 |
| Large / Regular | 17 15/16 | 22 ½ | 20 | 26 ½ | 28 |
| Large / Long | 17 15/16 | 22 ½ | 21 | 28 | 30 |
| Large / X-Long | 17 15/16 | 22 ½ | 22 | 29 ½ | 32 |
| X-Large / Short | 18 11/16 | 24 ½ | 20 | 25 ½ | 26 |
| X-Large / Regular | 18 11/16 | 24 ½ | 21 | 27 | 28 |
| X-Large / Long | 18 11/16 | 24 ½ | 22 | 28 ½ | 30 |
| X-Large / X-Long | 18 11/16 | 24 ½ | 23 | 30 | 32 |
| XX-Large / Short | 19 7/16 | 26 ½ | 21 | 26 | 26 |
| XX-Large / Regular | 19 7/16 | 26 ½ | 22 | 27 ½ | 28 |
| XX-Large / Long | 19 7/16 | 26 ½ | 23 | 29 | 30 |
| XX-Large / X-Long | 19 7/16 | 26 ½ | 24 | 30 ½ | 32 |
| Tolerances | -½ +1 | -½ + 1 | ± ½ | ± 5/8 | ± 1 |

E - The chest width shall be measured from folded edge to folded edge at underarm seam including knit inserts.
F - The hip shall be measured from folded edge to folded edge at bottom of knit inserts.
G - The center back length shall be measured from center back neck edge to waist seam along edge of zipper.
H - The sleeve length shall be measured along folded edge from neck edge of the garment to sleeve bottom. (Does not include the neck seal.)
J - The leg inseam length shall be measured along inner leg seam from middle of zipper to bottom of leg.



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Table V. Type IV - CWU-86/P finished garment dimensions (inches).

| SIZE | CHEST WIDTH<br>A | SLEEVE LENGTH<br>B | FRONT TORSO LENGTH<br>C | INSEAM LENGTH<br>D | YOKE WIDTH<br>K | CALF POCKET BOX DEPTH<br>L | ZIPPER FLAP LENGTH<br>M | COLLAR LENGTH<br>N |
|---|---|---|---|---|---|---|---|---|
| Small / Short | 21 ¼ | 22 ¼ | 22 ½ | 30 ½ | 13 ¾ | 2 ¼ | 9 ¼ | 18 |
| Small / Regular | 21 ¼ | 23 ¾ | 24 ½ | 33 | 13 ¾ | 2 ¼ | 9 ¼ | 18 |
| Small / Long | 21 ¼ | 24 3/8 | 26 | 34 ½ | 13 ¾ | 2 ¼ | 9 ¼ | 18 |
| Medium / Short | 23 | 22 3/8 | 23 ¾ | 30 ½ | 15 ½ | 2 ¼ | 9 7/8 | 18 ½ |
| Medium / Regular | 23 | 23 7/8 | 25 ¾ | 33 | 15 ½ | 2 ¼ | 9 7/8 | 18 ½ |
| Medium / Long | 23 | 24 7/8 | 27 ¼ | 34 ½ | 15 ½ | 2 ¼ | 9 7/8 | 18 ½ |
| Large / Short | 25 | 22 5/8 | 25 | 30 ½ | 17 ½ | 2 ¼ | 10 ½ | 19 ¼ |
| Large / Regular | 25 | 24 1/8 | 27 | 33 | 17 ½ | 2 ¼ | 10 ½ | 19 ¼ |
| Large / Long | 25 | 25 1/8 | 25 ½ | 34 ½ | 17 ½ | 2 ¼ | 10 ½ | 19 ¼ |
| X-Large Short | 27 ½ | 23 | 26 ¼ | 30 ½ | 20 | 2 ¼ | 11 1/8 | 20 |
| X-Large Reg | 27 ½ | 24 ½ | 28 ¼ | 33 | 20 | 2 ¼ | 11 1/8 | 20 |
| X-Large Long | 27 ½ | 25 ½ | 29 3/4 | 34 ½ | 20 | 2 ¼ | 11 1/8 | 20 |
| Tolerance | ± 1 | ± 5/8 | ± 1 | ± 1 | ± 3/8 | ± 1/8 | ±¼ | ±3/8 |

A - The chest width shall be measured above the zipper from edge to edge at the underarm seam.
B - The sleeve length shall be measured along the inseam sleeve from the armscye seam to the finished edge of the cuff.
C - The front torso length shall be measured along the center front seam from the edge of the neck seal seam to the edge of the material at the crotch. The measuring tape shall be placed to the right side of the relief portal zipper.
D - The inseam shall be measured along the seam from the edge of the crotch seam to the bottom edge of the leg.
K - The yoke cross back width shall be measured along the yoke stitchline, between armscye finished edges.
L - The calf pocket box depth shall be measured with the lower, rear corner of the calf pocket pulled out to its fullest volume. Measure the depth.
M - The overvest zipper flap length shall be measured from the topmost folded edge of the flap to its termination in the yoke attachment seam.
N - The collar length shall be measured with the overvest unzipped and the coverall neckline tucked into coverall body. Measurement of the collar shall be approximately 1 inch above and parallel to the neckline seam.



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TABLE VI. Type V - CWU-87/P finished garment dimensions (inches).

| SIZE | CHEST E | HIP F | CENTER BACK LENGTH G | SLEEVE LENGTH W/CUFF H | INSEAM LENGTH J | YOKE ACROSS BACK K | CALF POCKET DEPTH L | ZIPPER FLAP LENGTH M | COLLAR LENGTH N |
|---|---|---|---|---|---|---|---|---|---|
| X-Small / Short | 37 ½ | 33 | 16 | 27 ½ | 27 ½ | 15 ½ | 2 ¼ | 9 ½ | 16 ¼ |
| X-Small / Regular | 37 ½ | 33 | 17 | 29 | 29 ½ | 15 ½ | 2 ¼ | 9 ½ | 16 ¼ |
| X-Small / Long | 37 ½ | 33 | 18 | 30 ½ | 31 ½ | 15 ½ | 2 ¼ | 9 ½ | 16 ¼ |
| X-Small / X-Long | 37 ½ | 33 | 19 | 32 | 33 ½ | 15 ½ | 2 ¼ | 9 ½ | 16 ¼ |
| Small / Short | 39 | 37 | 17 | 28 | 27 ½ | 16 | 2 ¼ | 10 | 17 ¼ |
| Small / Regular | 39 | 37 | 18 | 29 ½ | 29 ½ | 16 | 2 ¼ | 10 | 17 ¼ |
| Small / Long | 39 | 37 | 19 | 31 | 31 ½ | 16 | 2 ¼ | 10 | 17 ¼ |
| Small / X-Long | 39 | 37 | 20 | 32 ½ | 33 ½ | 16 | 2 ¼ | 10 | 17 ¼ |
| Medium / Short | 42 ½ | 41 | 18 | 28 ½ | 27 ½ | 16 ½ | 2 ¼ | 10 ½ | 18 3/8 |
| Medium / Regular | 42 ½ | 41 | 19 | 30 | 29 ½ | 16 ½ | 2 ¼ | 10 ½ | 18 3/8 |
| Medium / Long | 42 ½ | 41 | 20 | 31 ½ | 31 ½ | 16 ½ | 2 ¼ | 10 ½ | 18 3/8 |
| Medium / X-Long | 42 ½ | 41 | 21 | 33 | 33 ½ | 16 ½ | 2 ¼ | 10 ½ | 18 3/8 |
| Large / Short | 46 | 45 | 19 | 29 | 27 ½ | 17 | 2 ¼ | 11 ¼ | 19 |
| Large / Regular | 46 | 45 | 20 | 30 ½ | 29 ½ | 17 | 2 ¼ | 11 ¼ | 19 |
| Large / Long | 46 | 45 | 21 | 32 | 31 ½ | 17 | 2 ¼ | 11 ¼ | 19 |
| Large / X-Long | 46 | 45 | 22 | 33 ½ | 33 ½ | 17 | 2 ¼ | 11 ¼ | 19 |
| X-Large / Short | 47 ½ | 49 | 20 | 24 ½ | 27 ½ | 17 ½ | 2 ¼ | 12 1/8 | 19 ¾ |
| X-Large / Regular | 47 ½ | 49 | 21 | 31 | 29 ½ | 17 ½ | 2 ¼ | 12 1/8 | 19 ¾ |
| X-Large / Long | 47 ½ | 49 | 22 | 32 ½ | 31 ½ | 17 ½ | 2 ¼ | 12 1/8 | 19 ¾ |
| X-Large / X-Long | 47 ½ | 49 | 23 | 34 | 33 ½ | 17 ½ | 2 ¼ | 12 1/8 | 19 ¾ |
| XX-Large / Short | 49 | 53 | 21 | 30 | 27 ½ | 18 | 2 ¼ | 13 | 20 ½ |





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TABLE VI. Type V - CWU-87/P finished garment dimensions (inches) - Continued.

| SIZE | CHEST<br>E | HIP<br>F | CENTER BACK LENGTH<br>G | SLEEVE LENGTH W/CUFF<br>H | INSEAM LENGTH<br>J | YOKE ACROSS BACK<br>K | CALF POCKET DEPTH<br>L | ZIPPER FLAP LENGTH<br>M | COLLAR LENGTH<br>N |
|---|---|---|---|---|---|---|---|---|---|
| XX-Large / Regular | 49 | 53 | 22 | 31 ½ | 29 ½ | 18 | 2 ¼ | 13 | 20 ½ |
| XX-Large / Long | 49 | 53 | 23 | 33 | 31 ½ | 18 | 2 ¼ | 13 | 20 ½ |
| XX-Large / X-Long | 49 | 53 | 24 | 34 ½ | 33 ½ | 18 | 2 ¼ | 13 | 20 ½ |
| Tolerances | + 1/-½ | + 1/-½ | ± ½ | ± 5/8 | ± 1 | ± 3/8 | ± 1/8 | ± ¼ | ± 3/8 |

E - The chest shall be measured from folded edge to folded edge at underarm seam including knit inserts. Double the measurement.
F - The hip shall be measured from folded edge to folded edge at bottom of knit inserts. Double the measurement.
G - The center back length shall be measured from center back neck edge to waist seam along edge of zipper.
H - The sleeve length shall be measured along topmost folded edge from neck edge of the garment to sleeve bottom. (Does not include the neck seal.)
J - The leg inseam length shall be measured along inner leg seam from middle of zipper to bottom of leg.
K - The yoke cross back width shall be measured along the yoke stitchline, between armscye finished edges.
L - The calf pocket box depth shall be measured with the lower, rear corner of the calf pocket pulled out to its fullest volume. Measure the depth.
M - The overvest zipper flap length shall be measured from the topmost folded edge of the flap to its termination in the yoke attachment seam.
N - The collar length shall be measured with the overvest unzipped and the coverall neckline tucked into coverall body. Measurement of the collar shall be 1 inch above and parallel to the neckline seam.




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3.14 Performance Requirements.

3.14.1 End item coverall leakage. The anti-exposure coveralls shall not leak. A wax pencil shall be used for all marking and there shall not be any evidence of construction or material failure in any respect when tested as specified in 4.5.3. The inspection form shall be filled out (see figures 9, 10, 11, and 12).

3.15 Protective storage requirements. Each coverall shall be neatly folded in accordance with figure 13 using tissue paper in the fold areas to preclude sharp creases from forming. The folded coveralls shall be inserted into a waterproof bag that is double folded at the end and sealed using two-inch general purpose masking tape. The bag shall not be heat sealed.

3.16 Workmanship. The finished coveralls shall be uniform in quality and free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2)

b. First article inspection (see 4.3)

c. Conformance inspection (see 4.4)

4.2 Qualification inspection. The qualification inspection of the coveralls shall consist of the examinations and tests specified in 4.5.1 through 4.5.3.

4.2.1 Qualification samples. The qualification samples shall consist of three coveralls of types I, II or IV (see 3.1); one each of Size 1, 6, and 12 or three coveralls of types III or V; one each size Small/Regular, Medium/Short, Large/Long.

4.3 First article inspection. First article inspection shall consist of the examinations and tests specified in 4.5.1 through 4.5.3.

4.3.1 First article samples. The number of samples for first article inspection shall be specified in the contract (see 6.2). The first article samples shall include one coverall of each size ordered. The sample unit shall be one coverall and the lot size shall be expressed in units of coveralls.

4.4 Conformance inspection. Conformance inspection shall consist of the examinations and tests specified in 4.5.1 through 4.5.3.



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4.4.1 Conformance inspection samples. Sampling for conformance inspection shall be performed in accordance with ASQ-Z1.4 (see 6.2). The sample unit shall be one coverall and the lot size shall be expressed in units of coveralls.

4.5 Examinations and tests.

4.5.1 Material and component inspection. Materials and components shall be examined and tested in accordance with table VII.

TABLE VII. Materials and components examinations and tests.

| MATERIAL OR COMPONENT | REQUIREMENT PARAGRAPH | EXAMINATIONS AND TESTS |
|---|---|---|
| Laminated cloth (Types I, II and III) | 3.4.1 | All examinations and tests specified in MIL-C-85637 |
| Laminated cloth (Types IV and V) | 3.4.2 | All examinations and tests specified in MIL-DTL-32149 |
| Raschel knit | 3.4.3 | All examinations and tests specified in MIL-C-85101 |
| Aramid cloth (Types IV and V) | 3.4.4 | All examinations and tests specified in MIL-C-83429 |
| Heat seal cloth (Types IV and V) | 3.4.5 | All examinations and tests specified in 1370AS502 |
| Zippers | 3.4.6 | All examinations and tests specified in 3769AS103 and A-A-55634 as applicable |
| Fastener tape (Types III, IV and V) | 3.4.7 | All examinations and tests specified in A-A-55126 |
| Thread | 3.4.8 | All examinations and tests specified in A-A-50195 |
| Snap Hook | 3.4.9 | All examinations and tests specified in NASM43770 |

4.5.2 End item inspection. The end item shall be visually examined, measured and tested as specified herein.

4.5.2.1 Classification of end item defects. The end items shall be examined for the defects listed in table VIII.



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TABLE VIII. Classification of end item defects.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | CRITICAL | MAJOR | MINOR |
| GENERAL | Any hole, scissor or knife tear, mend, burn, or weakening defect such as multiple floats, slubs, skips, needle chew or abraded area. | | 101 | |
| | Any evidence of fabric delamination. | | 102 | |
| | Any spot or stain (compound, oil, dirt, incl marks) clearly visible. 1/ | | | 201 |
| | Color or shade of any component not as specified. | | | 202 |
| | Any thread not trimmed to 1/16 inch and not covered by heat sealed seam tape. | | 103 | |
| | Any thread not trimmed to 1/16 inch or thread scraps not removed. | | | 203 |
| | Any neck seal attachment seam not covered with the specified tape. | | 104 | |
| | Zipper not covered properly or not covered with the specified tape. | | 105 | |
| | Any damaged tape edges greater than 1/16 inch from the outside edge. | | | 204 |
| | More than three, two-inch diameter mending patches per coverall. | | 106 | |
| | Inspection form missing or not completely filled out. | | 107 | |
| | Marking of inside of suit with utensil other than a wax marking pencil. | | 108 | |
| | Material and components more than 18 months old. | 1 | | |



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TABLE VIII. Classification of end item defects - Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | CRITICAL | MAJOR | MINOR |
| ZIPPERS | Any part of a zipper bent, broken, or otherwise defective. | | 109 | |
| | Zipper not closing as specified. 2/ | | 110 | |
| | Length of zipper not as specified. | | 111 | |
| | Entry or portal zipper excessively lubricated or not lubricated. | | | 205 |
| COMPONENTS AND ASSEMBLY | Any defective component or defect that will affect the form, fit, and function of the assembly. | | 112 | |
| | Any component part not as specified, omitted, or operation improperly performed. | | 113 | |
| | Any component part not cut in accordance with the patterns. | | 114 | |
| HEAT SEALING | Any seam tape not located as specified | | 115 | |
| | Any stitching not covered by seam tape. | | 116 | |
| | Any splice taping not overlapped ¾ inch. | | 117 | |
| SEAMS AND STITCHING | Any seam or attachment of any component twisted, puckered, pleated, or caught in any unrelated operation or stitching that is not properly forced out or contained in a fold more than 1/8 inch. | | 118 | |
| | End of stitching not securely backstitched for at least ½ inch when not caught in other seams or stitching. | | 119 | |
| | Thread breaks, skips, and run-offs not securely overstitched for at least ½ inch. | | 120 | |
| | Any stitching irregular or unevenly gauged (greater than 50% of the seam length or 4 inches, whichever is less). | | 121 | |



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TABLE VIII. Classification of end item defects - Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | CRITICAL | MAJOR | MINOR |
| SEAMS AND STITCHING Continued | Not specified seam or stitch type. | | 122 | |
| | Loose tension resulting in a loose seam or tight tension resulting in breaking of stitches when normal pull is applied 3/. | | 123 | |
| | Stitches per inch - one or two stitches more or less than specified (to be scored only when condition exists on 25% of the seam or more). | | | 206 |
| | Stitches per inch more than two stitches more or less than specified (to be scored only when condition exists on 25% of the seam or more). | | 124 | |
| | Any open seam 4/. | | 125 | |
| LABEL AND TAG | Label/tag missing or insecurely attached. | | 126 | |
| | Size/information on label missing, incorrect, or illegible. | | 127 | |
| | Point size on label incorrect. | | | 207 |
| FINISHED GARMENT | Coverall not properly folded. | | | 208 |
| | Coverall not preserved as specified. | | | 209 |
| POCKET ZIPPER COVERINGS (Type II only) | Edges of zipper coverings overlapping more than 1/16 inch or gapped more than 1/16 inch, when zipper is closed. | | | 210 |
| | Beading omitted in zipper covering. | | 128 | |
| | Beading not sewn taut in zipper covering. | | | 211 |



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TABLE VIII. Classification of end item defects - Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | CRITICAL | MAJOR | MINOR |
| BARTACK (Type II only) | Any bartack omitted, loose, misplaced, or not serving intended purpose. | | 129 | |
| | Bartack omitted, loose, misplaced, or not serving intended purpose on multiple pencil compartment pocket: | | | |
| | one bartack | | | 212 |
| | two or more bartacks | | 130 | |
| | Length of any bartack on multiple pencil compartment pocket not as specified. | | | 213 |
| POCKET (Type II only) | Any breast, thigh, or lower leg pocket out of horizontal or vertical alignment: | | | |
| | more than 3/8 inch, but not more than ½ inch | | | 214 |
| | more than ½ inch | | 131 | |
| | Any breast, thigh, or lower leg pocket poorly shaped or not uniform in size and shape. | | | 215 |
| | Pencil compartment pocket not on left sleeve. | | 132 | |
| | Pencil pocket protective flap omitted. | | 133 | |
| | One pencil compartment omitted. | | | 216 |
| | Omitted, misplaced, incorrect size | | 134 | |
| | Omitted, does not mate | | 135 | |
| POCKET (Type III and V) | Omitted | | 136 | |



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TABLE VIII. Classification of end item defects - Continued.

| EXAMINE | DEFECT | CLASSIFICATION | | |
|---|---|---|---|---|
| | | CRITICAL | MAJOR | MINOR |
| HOOK AND LOOP FASTENER TAPE (Type III, IV, and V ) | Omitted, does not align | | 137 | |
| ZIPPER THONG (Type III and V) | Omitted | | 138 | |
| | Dimensions not as specified | | 139 | |
| END ITEM DIMENSIONS | Any dimension not within specified tolerance | | 140 | |
| | Any construction dimension that affects form, fit, or function | | 141 | |
| | Any construction dimension that does not affect form, fit, or function | | | 217 |

1/ Clearly visible shall be interpreted as meaning clearly visible at an examination distance of 3 feet.
2/ The zippers shall be checked for proper function by opening and closing each zipper not less than three times along its full length.
3/ Puckering is evidence of tight tension or gathering of the material. When puckering is evident and is caused by the gathering of the material, the stitching shall be inspected for tight tension and breaking of stitches by exerting normal pull in the lengthwise direction of the stitching, by pulling the material taut to straighten out the seam. Puckering due to the gathering of the material shall be classified and scored as a seaming defect.
4/ A seam shall be considered open whether one side or both sides are open.

4.5.2.2 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in 3.13 and tables II through VI, as applicable.

4.5.3 Coverall leakage. Using the apparatus specified in MIL-W-85635 and the procedures specified herein, each of the assembled coveralls shall be tested for conformance to the requirements of 3.14. The water leakage test shall be accomplished in one complete cycle from start to finish without interruption.

a. Fill the reservoir with approximately 30 gallons of clean, fresh water and add 60 milliliters (ml) ($^1/_4$ cup) of chlorine bleach. The bleach shall be added at the initial filling of the reservoir and at each subsequent refilling. Do not add more than 60 ml ($^1/_4$ cup) of bleach for



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each reservoir full of water. The reservoir shall be drained and the water replaced at least monthly or more often if the water becomes dirty, slimy, dank, or odoriferous.

b. Secure the water leakage inspection apparatus in the horizontal position with the platform open.

c. The garment to be inspected shall be turned inside out with the entrance and relief zippers, as applicable, fully closed. Lay the coverall on the bottom half of the platform, positioning the neck opening to the left and the entrance zipper facing up. Care shall be taken while inserting the coverall in the test apparatus to ensure that the garment is not pinched.

d. Close off the left wrist, neck, and each leg opening with a clamp. The openings shall be folded prior to clamping to avoid leakage at the openings.

e. Extend the right sleeve of the coverall through the opening provided in the side of the inspection apparatus and close the top gate. Secure all four latches.

f. While holding the fill hose and right sleeve, turn the pump on and fill the coverall to within 6 inches of the right sleeve wrist opening. Allow the trapped air to escape through the wrist opening. If the water does not flow from the filling hose within 5 seconds from the time the switch is turned to the "on" position, prime the pump by clicking the switch "off" and "on" several times.

g. Turn the pump off and rotate the inspection apparatus to a vertical position and secure the platform. If necessary, turn the pump on again and refill the coverall to within 6 inches of the right sleeve wrist opening.

h. Turn the pump off, remove hose and clamp off right sleeve wrist opening in a similar manner as described in 4.5.3d.

i. While the inspection apparatus is in the vertical position and locked in place, visually inspect the coverall front and back for leakage. If water appears on the coverall surface, blot area dry with an absorbent material, such as a paper towel, and observe whether the water reappears. If water does not reappear within 10 seconds, it will not be considered a leak.

j. If the water reappears, mark the suspect area with a wax marking pencil for repair. Do not use any other kind of pencil for marking. If minor leakage is attributable to the stitched seams exposed on the outside perimeter of the zipper area, it is acceptable to apply a thin layer of a neoprene base adhesive to the inside of the rubber covered zipper tape.

k. Rotate the lower end of the inspection platform forward until the lower slidebolt is aligned with the lower edge of the inspection frame. Move slidebolt over to rest on lower edge



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of inspection frame as an aid in holding this position. Inspect the lower side seam area and inseam area of the uppermost leg for leaks. Mark suspect areas as specified in 4.5.3i and 4.5.3j.

l. Rotate the upper end of the inspection platform forward 180 degrees until upper slidebolt is aligned with the lower edge of the inspection frame. Move slidebolt over to rest on the lower edge of inspection frame as an aid in holding this position. Inspect and mark lower side seam area and inseam area of uppermost leg as specified in 4.5.3i and 4.5.3j.

m. After marking all suspect areas for repair, rotate platform back to original horizontal position and lock in place. Remove sleeve clamp from right wrist and slowly drain water from garment.

n. Open platform, remove clamps, and drain any remaining water.

o. Turn coverall right side out.

p. Fully dry coverall. When using a dryer, set dryer at lowest temperature setting and dry coverall for a maximum of 20 minutes. When air drying, hang the coverall on a wooden hanger and allow to air dry. After hanging for a maximum of 4 hours, check coverall for dryness.

q. Ensure that all areas requiring patching are clearly marked.

r. Determine repairability of a coverall by patching (see 3.12.1).

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military services system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coveralls are military unique because they are used for continuous wear by military aircrew members on flights over water, in the event of immersion into water. The coveralls are intended to protect the wearer from exposure to cold water, wind, spray, and



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rain, and to protect against hypothermia. The coveralls are intended to be used in conjunction with waterproof wrist seals and socks fitted on an individual basis as applicable (see 3.6.1, 3.6.2, 3.6.3, 3.6.4, and 3.6.5).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.

b. Type, quantity desired (see 1.2.1).

c. National stock number (see 3.5.1).

d. Applicable Government patterns and drawings, including revisions.

e. Whether first article and conformance inspection is required (see 4.3 and 4.4).

f. Number of first article inspection samples (see 4.3.1).

g. Name and address of the first article inspection facility; and the name and address of the Government activity responsible for conducting the first article inspection program.

h. Packaging required (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-85633 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from: Commander, Naval Air Systems Command, Aviation Life Support Systems, Code 4.6.2.2, Building 2187 Suite 1242, 48110 Shaw Road, Unit 5, Patuxent River, MD 20670-1906.

6.3.1 Qualification samples. The samples should be forwarded to the test facility set forth in the letter of authorization to submit samples (see 6.3). The samples should be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
Coveralls, Flyer's, Anti-Exposure (Type as applicable)
Manufacturer's designation or number
Name of manufacturer
Submitted by (name) (date) for qualification inspection in accordance with the



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requirements of this document under authorization (reference authorizing letter and number) (see 6.3).

6.3.2 Retention. The retention of qualification listing should consist of periodic verification to determine compliance of the qualified coveralls with the requirements of this document. Periodic verification should be by certification unless otherwise specified by the activity responsible for the Qualified Products List and should be at intervals of not more than two years.

6.4 First article. When first article inspection (see 3.1) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3.1. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Patterns. To request patterns, contact the procuring activity issuing the invitation for bid.

6.5.1 Pattern numbers. The patterns listed in 3.8.1, table I have been identified with a NAVAIR drawing number for the purpose of configuration control and tracking of pattern changes. These numbers range from 3324AS181 - 3324AS192 for the type I coverall and 1370AS481 - 1370AS493 for the type II coverall. These numbers do not have any effect on manufacturing.

6.6 Sources of supply.

6.6.1 Entry and portal zippers. Omni-environmental barrier zippers meeting the requirements of 3.4.6.1 are available from the following source:

YKK (USA) Inc., c/o Diversified Marketing Group, 109 Forrest Avenue, Narberth, PA 19072, Vendor part number: TZ NC-81



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6.6.2 Pocket and overvest zippers. The following part numbers are applicable (see 3.4.6.3):

00I-07-IEF-C-05-N-26/4 for the pencil pocket zipper
00I-04-IEF-C-05-N-36/4 for the calf pocket
00I-04-IEF-C-05-N-lengths as specified for the overvest zippers

6.7 Shelf-life. This specification covers items where shelf-life is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order. The shelf-life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from *DoD 4140.27-M Shelf-life Management Manual,* or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: http://www.shelflife.hq.dla.mil/ .

6.8 Subject term (key word) listing.

Dry-Suit
Flight Clothing
Neck Seal
Sealing Machine, Heat
Shelf-life
Water Immersion
Wrist Seal

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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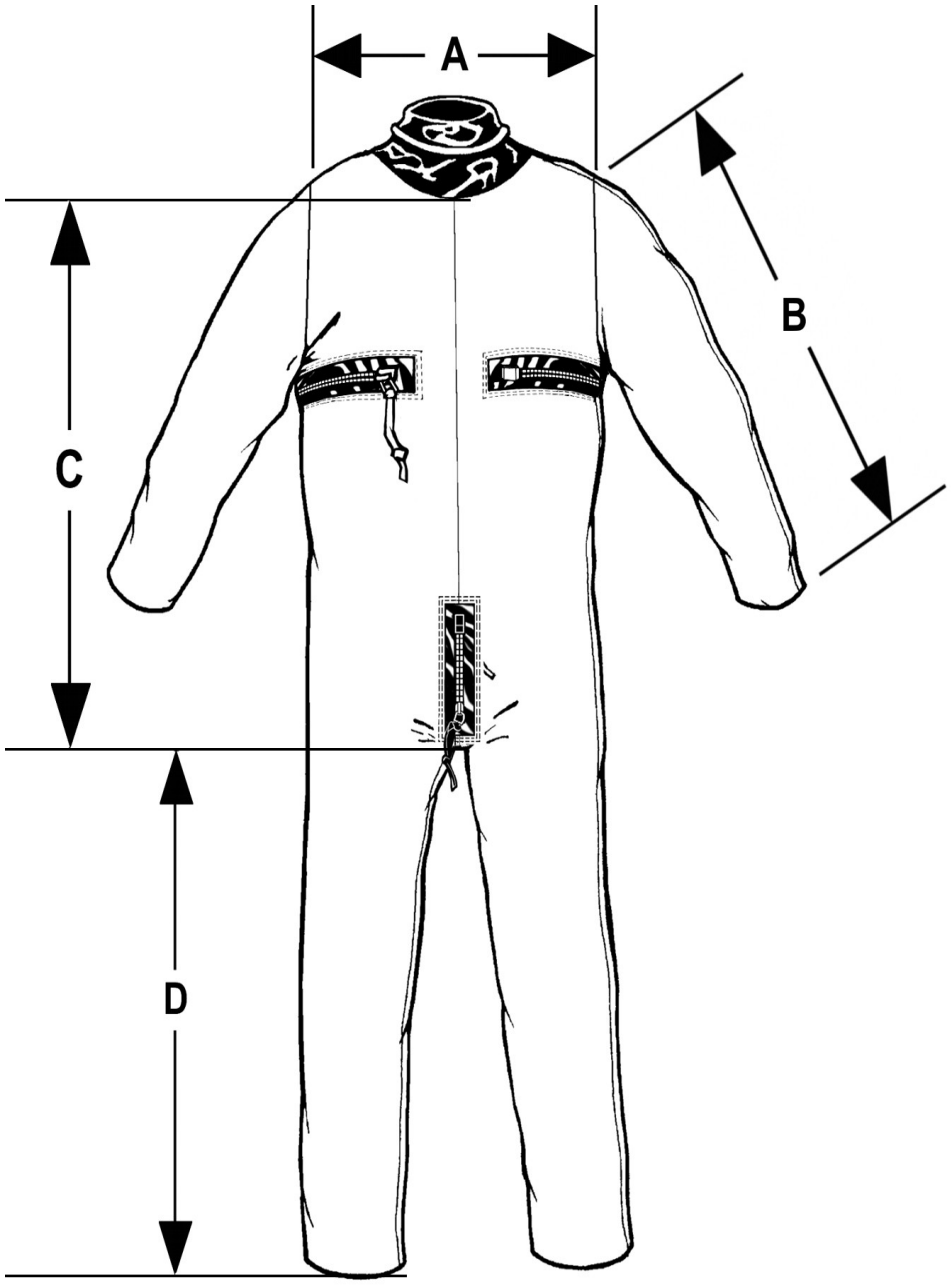



FIGURE 1. CWU-62B/P front



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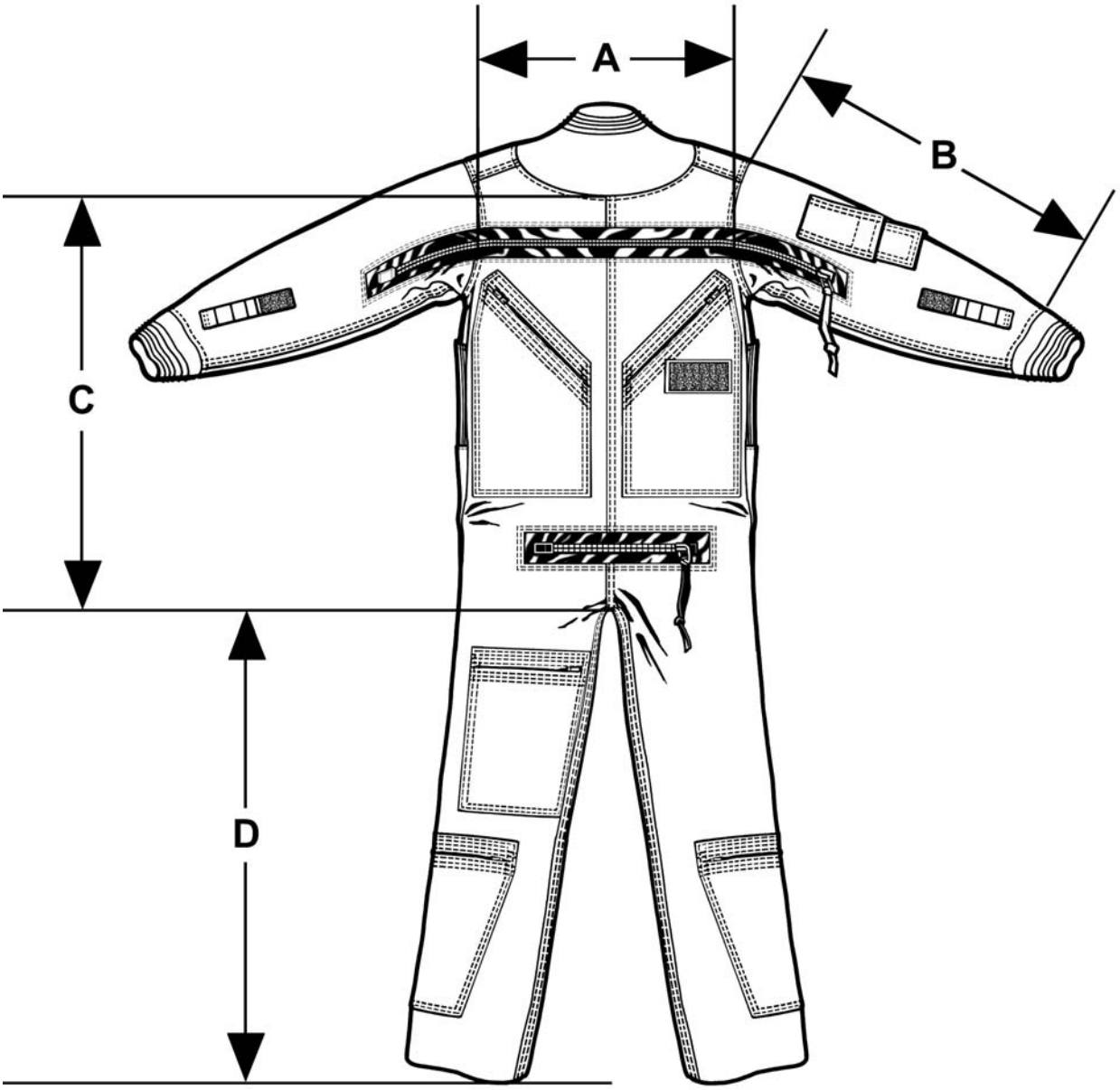



FIGURE 2. CWU-74/P



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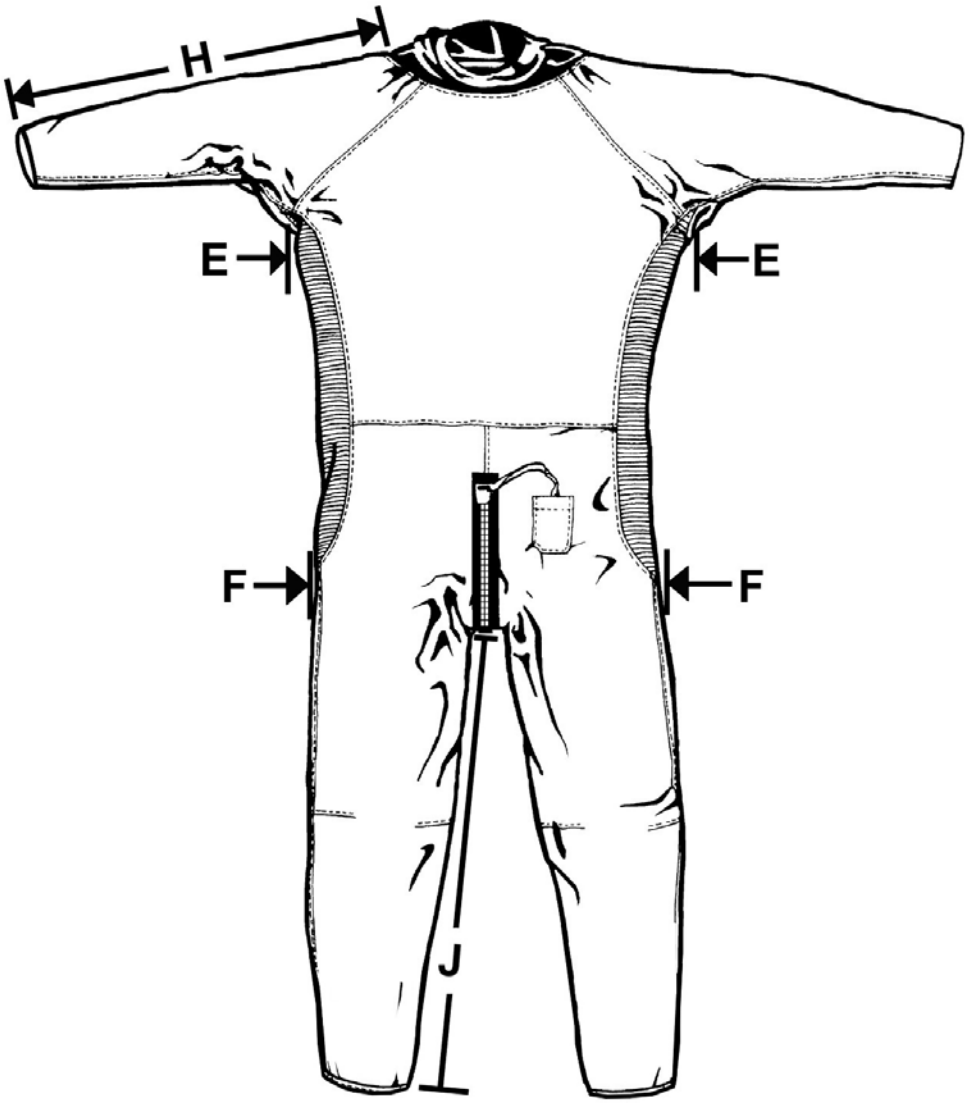



FIGURE 3. CWU-62C/P front.



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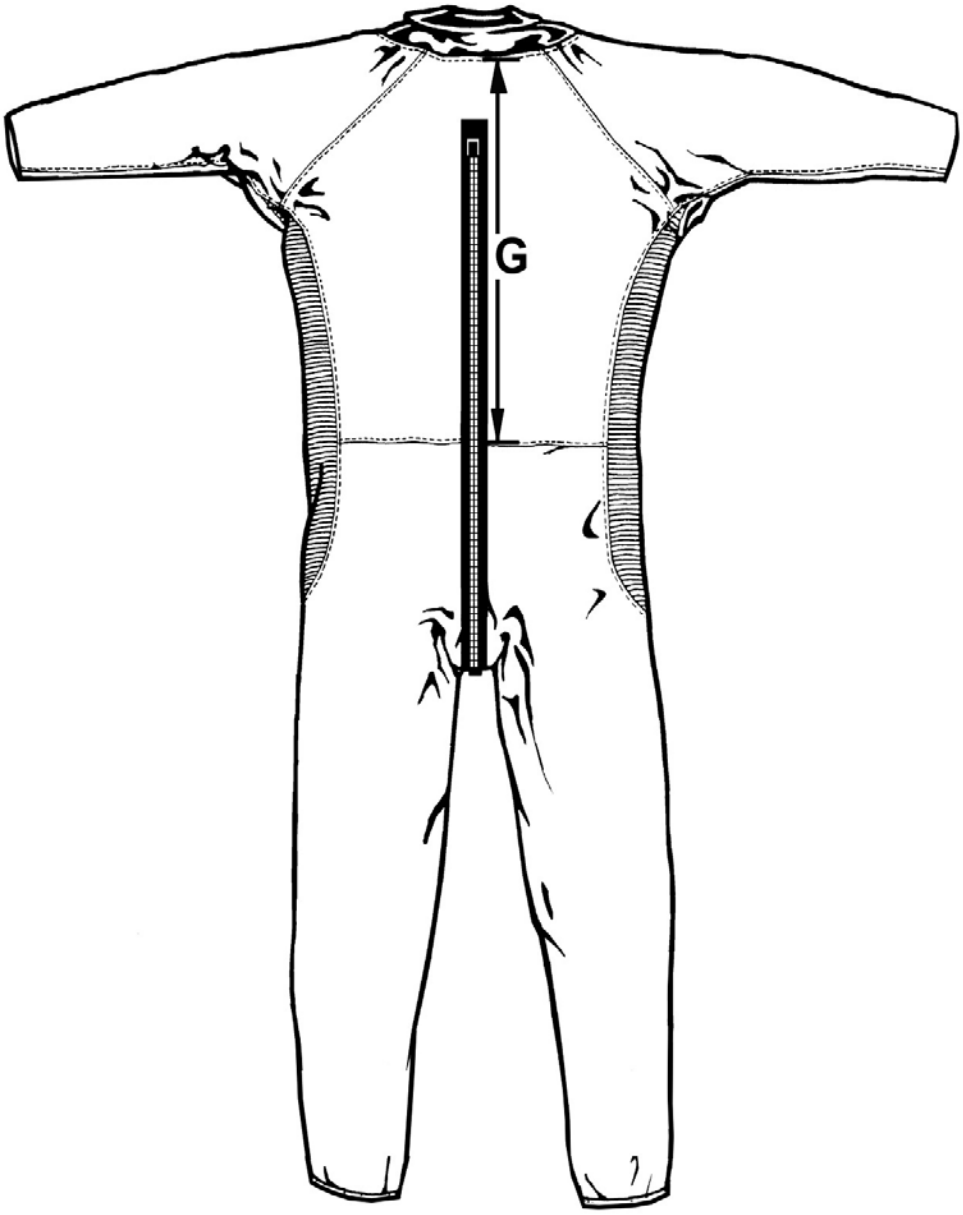



FIGURE 4. CWU-62C/P back



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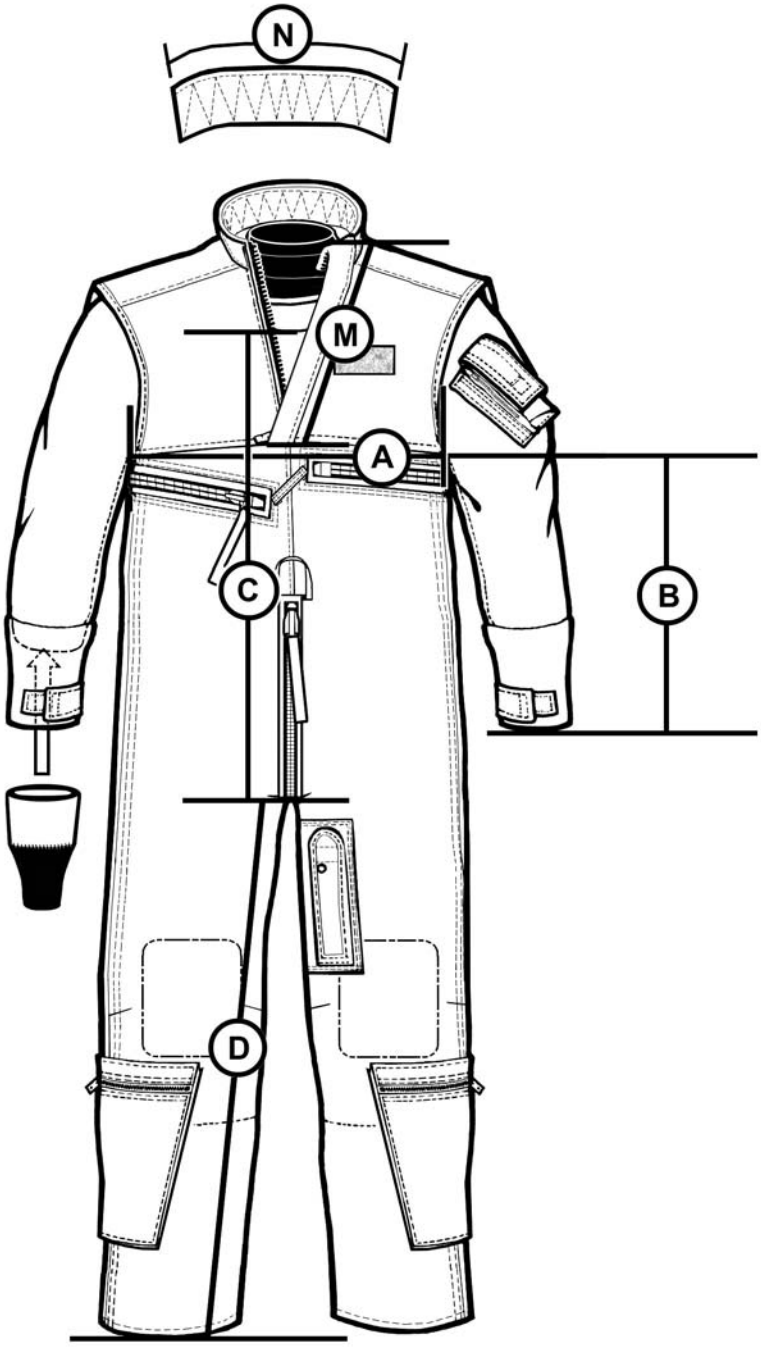



FIGURE 5. CWU-86/P front



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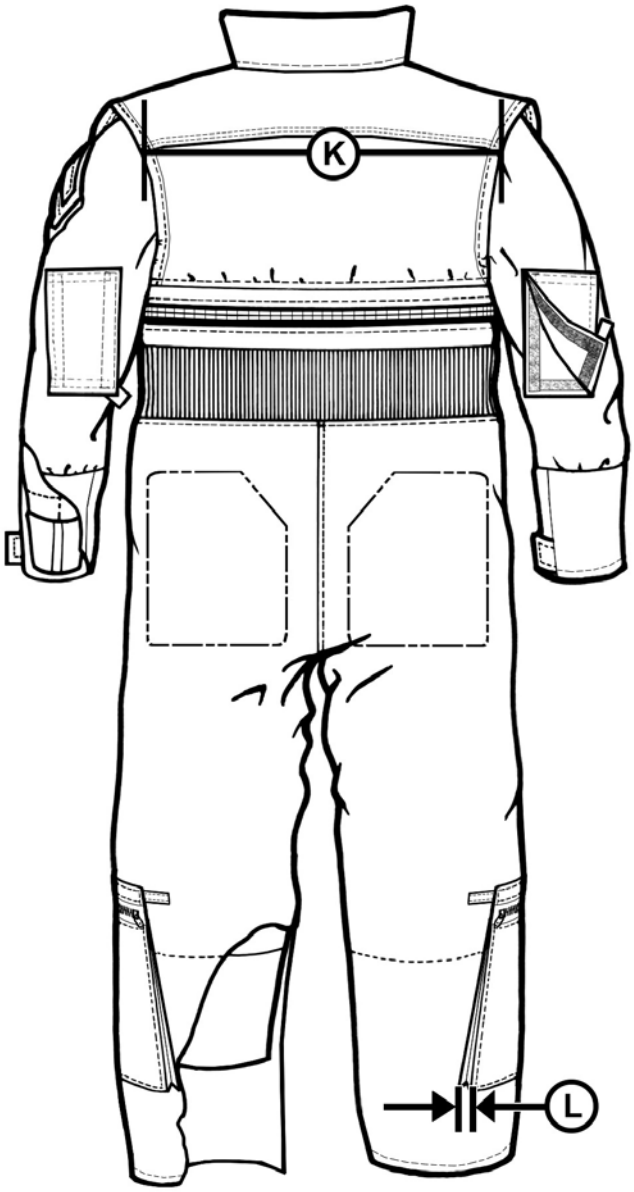



FIGURE 6. CWU-86/P back



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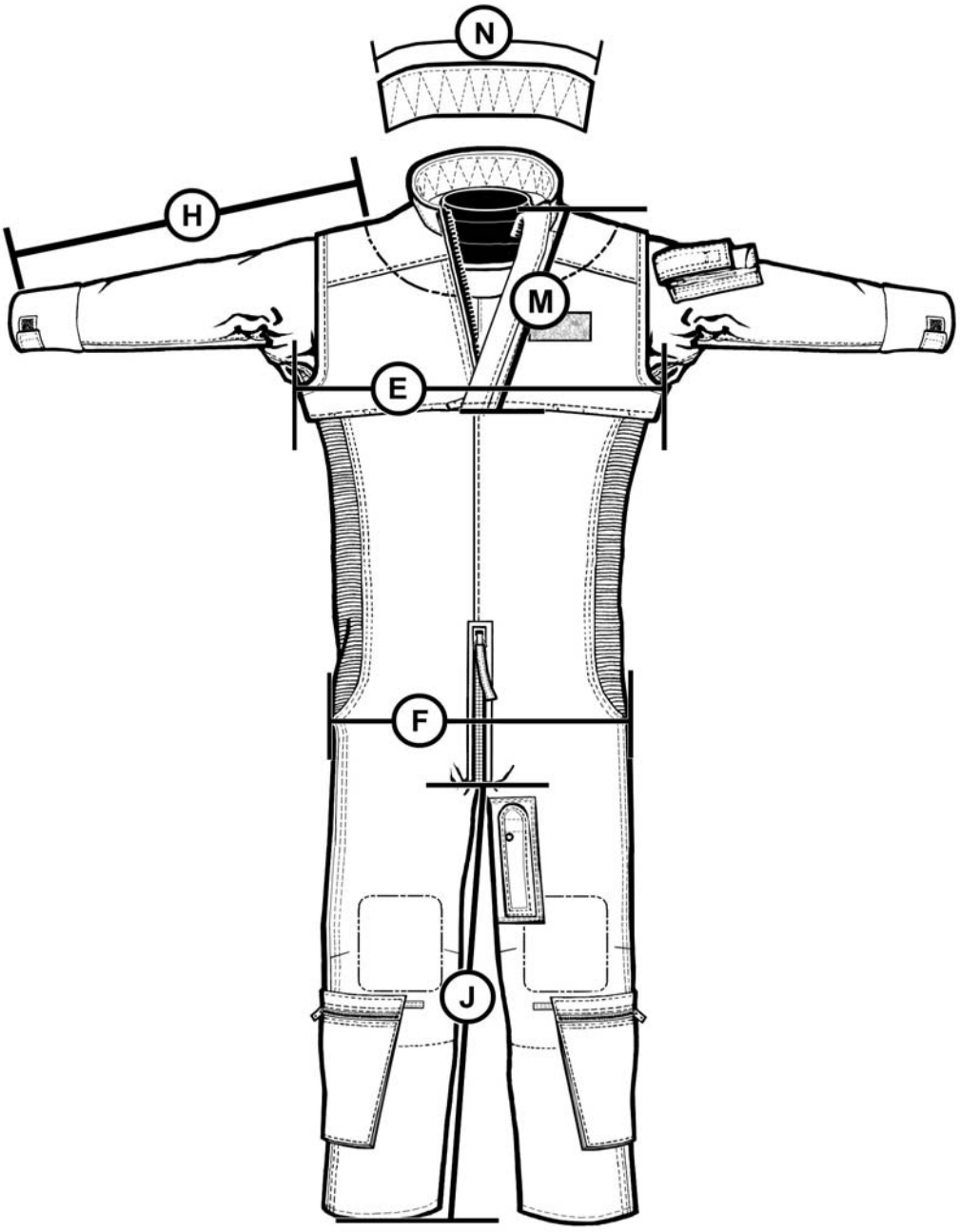



FIGURE 7. CWU-87/P front



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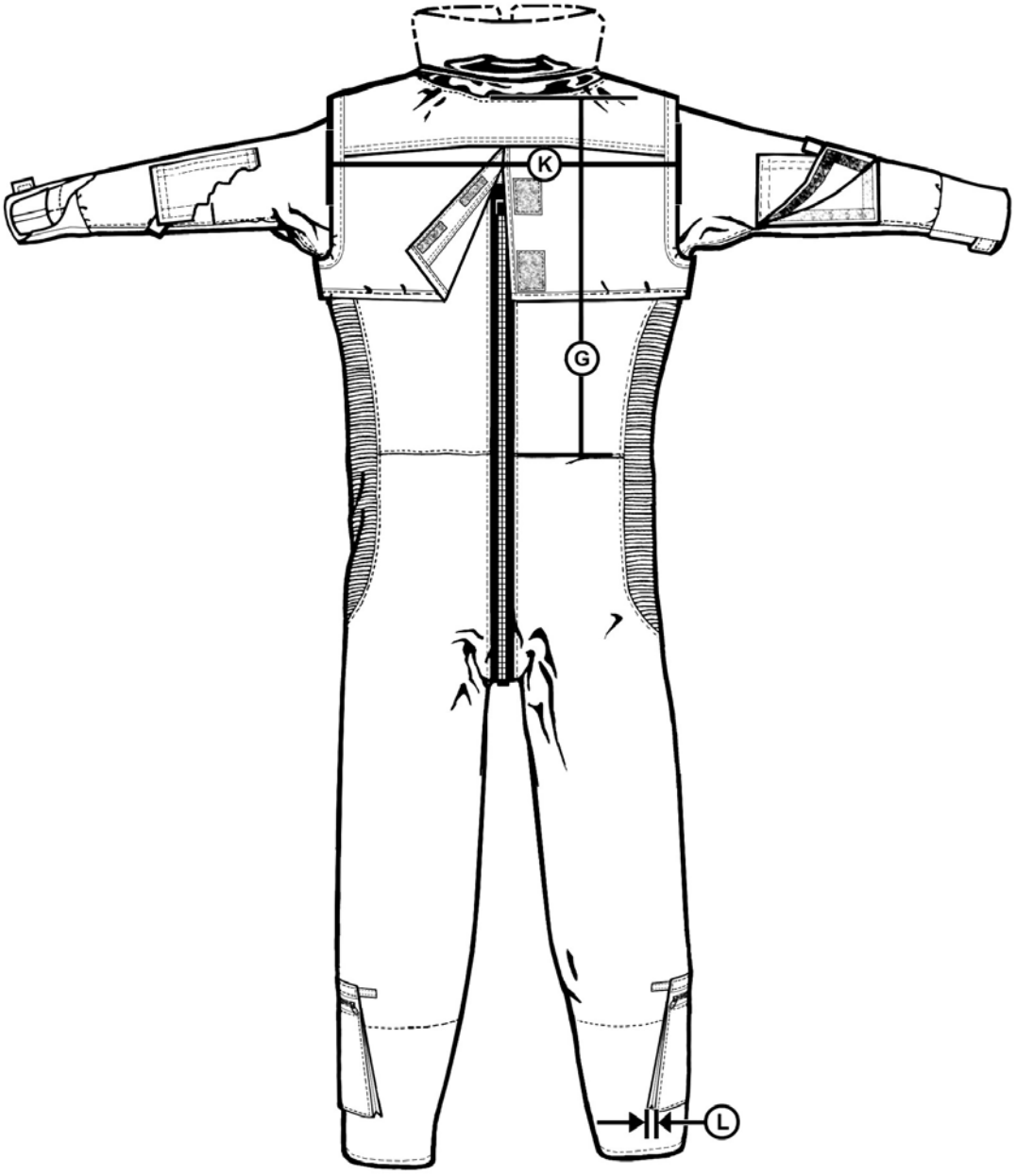



FIGURE 8. CWU-87/P back



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Name of Contractor:_____ Date:_____

Contract Number:_____ Coverall Serial Number: _____

No Leaks _____

Leaks (number) _____ Repaired (Date) _____

Passed: Yes_____ No _____ Inspected by:_____

Mark repair location on diagram below (suit is shown inside out):

FRONT BACK

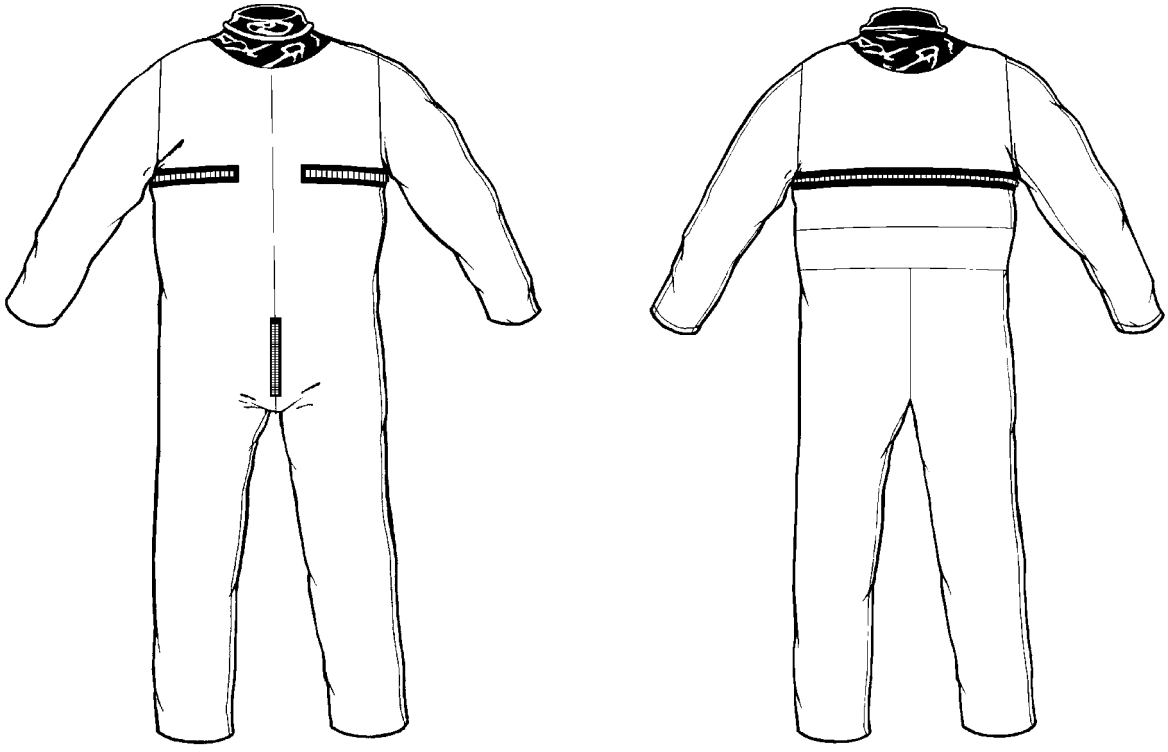


FIGURE 9. CWU-62B/P Anti-exposure coverall, patch location, and test information.



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Name of Contractor: _____ Date: _____

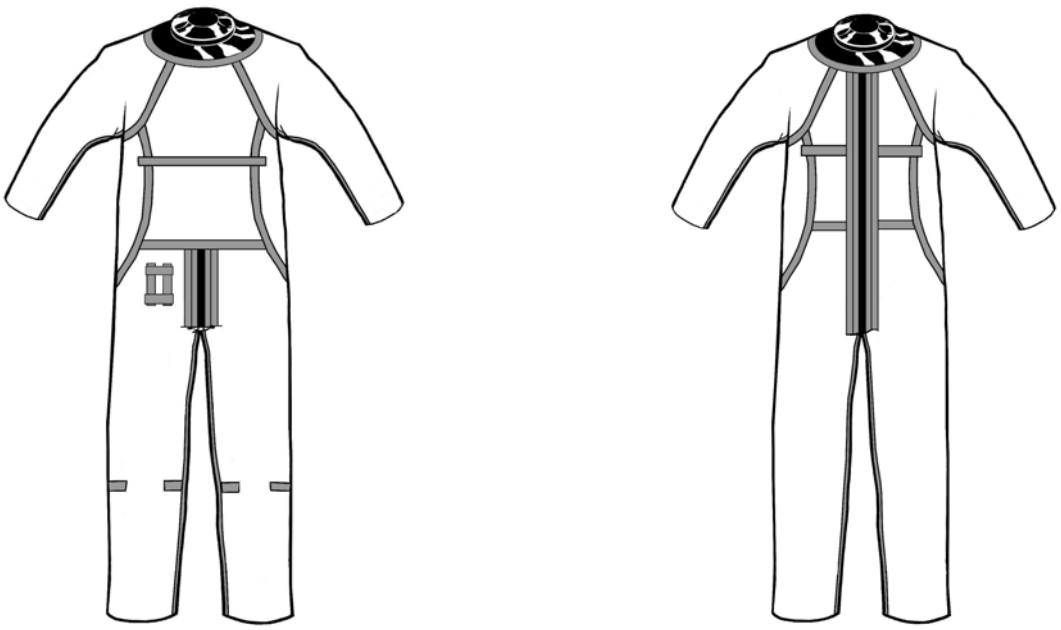
Contract Number:_____ Coverall Serial Number: _____

No Leaks _____

Leaks _____ Repaired (Date) _____

Ready For Issue:_____ Inspected by:_____

Mark repair location on diagram below (suit is shown inside out):



FRONT BACK

FIGURE 10. CWU-62C/P Anti-exposure coverall, patch location, and test information.



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Name of Contractor: _____ Date: _____

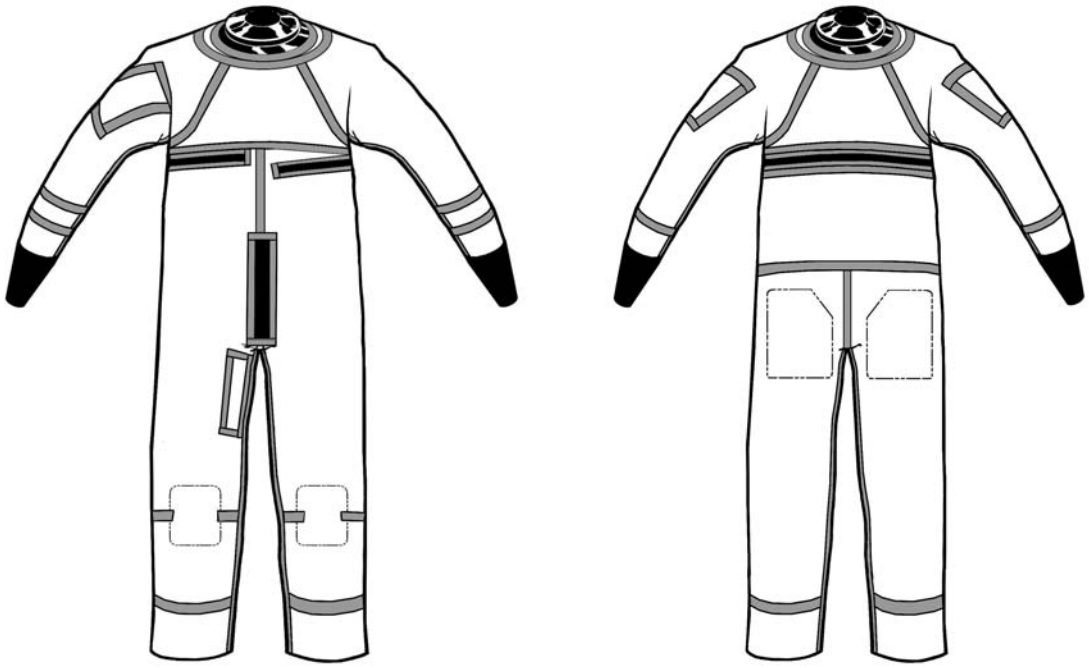
Contract Number: _____ Coverall Serial Number: _____

No Leaks: _____

Leaks (number): _____ Repaired (Date): _____

Passed: Yes_____ No_____ Inspected by: _____

Mark repair location on diagram below (suit is shown inside out):



FRONT BACK

FIGURE 11. CWU-86/P Ruggedized Anti-Exposure Coverall, Patch Location and Test Information



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Name of Contractor: _____ Date: _____

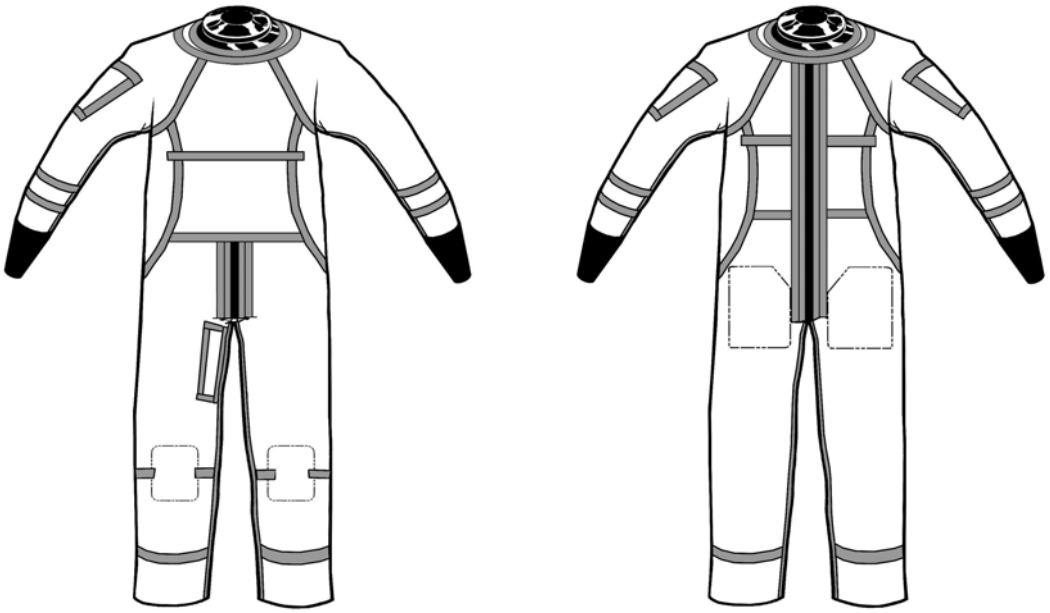
Contract Number: _____ Coverall Serial Number: _____

No Leaks: _____

Leaks (number): _____ Repaired (Date): _____

Passed: Yes_____ No_____ Inspected by: _____

Mark repair location on diagram below (suit is shown inside out):



FRONT BACK

FIGURE 12. CWU-87/P Ruggedized Anti-Exposure Coverall, Patch Location and Test Information



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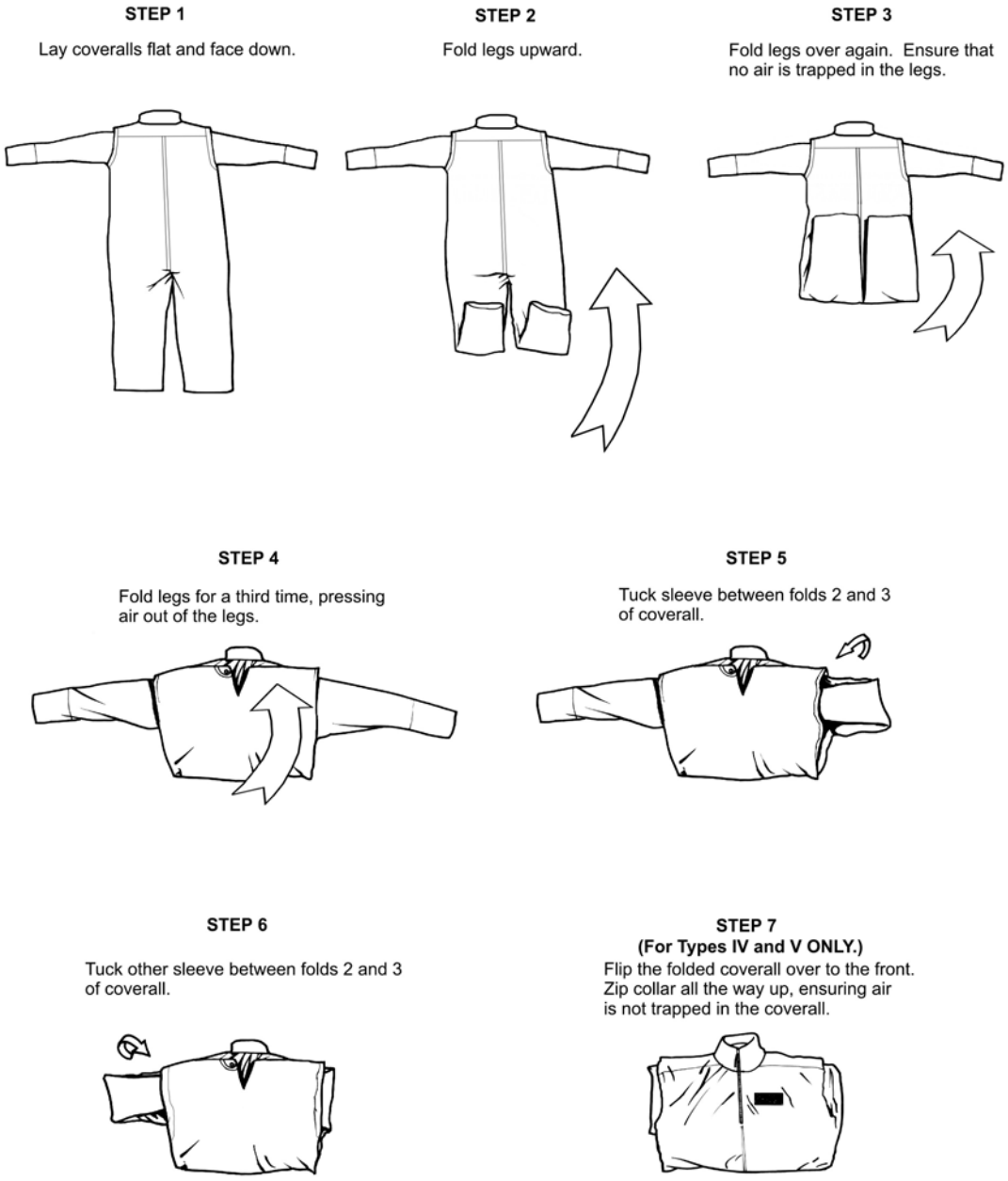



FIGURE 13. Coveralls folding procedures.



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Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA-CT

Review activities:
Navy-NU

Preparing Activity:
Navy - AS
(Project 8475-0016)

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